OBSOLETE

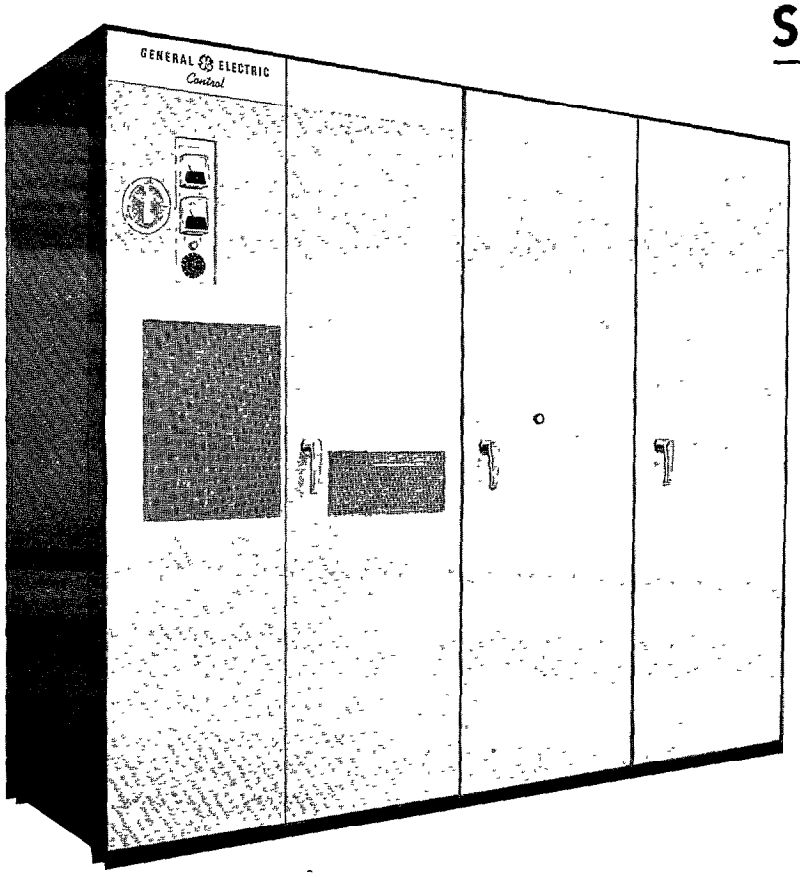
GEK-28544
use with
GEK-28543 (Section I)
GEK-28545 (Section III)
GEK-28546 (Section IV)

SILTROL* I

IC 3610 Integrated Static Conversion and Control Equipment for Adjustable-Speed Drives

*TRADEMARK OF GENERAL ELECTRIC CO.

INSTRUCTIONS
SECTION II



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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

GENERAL ELECTRIC

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SAFETY PRECAUTIONS

During installation and maintenance all power sources should be removed.

Adjustment or testing of energized circuits should be done only by authorized persons who are familiar with hazards involved.

ELEMENTARY DIAGRAMS

Because the Siltrol equipments are basically standard items the elementary diagrams are standardized with variable information being provided on data tables within the elementaries.

To completely describe a particular Siltrol there are four levels of elementary diagrams which are as follows:

LEVEL 1 - Requisition Drive Elementary

Within each requisition drive elementary there are two sheets which describe the Siltrol for that drive:

1) The Symbol Sheet consisting of a symbol functionally describing the Siltrol in block form only, but showing all incoming and outgoing interconnections in detail.

2) The Siltrol Power Convertor Design Variable Data Sheet showing all relevant data for that particular Siltrol. This is presented in the form of a table for each sheet of the Siltrol Power Convertor Elementary, showing all devices and functions, relevant catalog numbers, adjustments, and notes.

LEVEL 2 - Siltrol Power Convertor Elementary

The Level 2 elementary diagrams describe the Siltrol Power Convertor in modular form, with each module represented by a symbol block describing functionally how it works. All wires into each module are shown in detail. The elementary also includes data tables showing variable information. To cover all 6 types of Siltrol equipments produced, there are 6 versions of this elementary. But, only those required to cover the types of Siltrol equipment used in this requisition are included in the Instruction Book.

The six versions are:

Small HP - Non-reversing
Small HP - Reversing
Medium HP - Non-reversing
Medium HP - Reversing
Large HP - Non-reversing
Large HP - Reversing

The Level 2 diagrams are located in Section 2 of this Instruction Book.

LEVEL 3 - Siltrol Module Elementaries

The Siltrol equipment is divided into three modules:

1. Power Bridge Module
2. Firing Circuit Module.
3. Gating Module

1) This elementary shows the power bridge circuit in detail. As the power bridge is ordered by catalog number a complete breakdown of the number is given as part of the elementary.

2) The firing circuit is shown in detail except that each card in the firing circuit module is represented by a symbol which shows functionally how it works. A data sheet is also provided showing possible variations related to catalog number.

3) Similarly the gating module is shown in detail with each card represented by a symbol which shows functionally how it works. Again a data table is provided showing the possible variations related to catalog number.

The Level 3 elementary diagrams are located in Section 3 of the Instruction Book.

LEVEL 4 - Siltrol Card Elementaries

These elementaries describe the various cards used in the Siltrol equipment in detail and consist of:

1. Circuit diagrams
2. Symbol diagrams
3. Application Data

The Level 4 elementary diagrams are located in Section 4 of this Instruction Book.

FIRING CIRCUIT - DETAILED DESCRIPTION

To fully understand the operation of the Siltrol I equipment, a complete understanding of the firing circuit is essential. Fig. 4 shows the Level 2 symbol diagram for a small HP reversing firing circuit. The diagram describes in detail functionally how the firing circuit works.

It is not a circuit diagram.

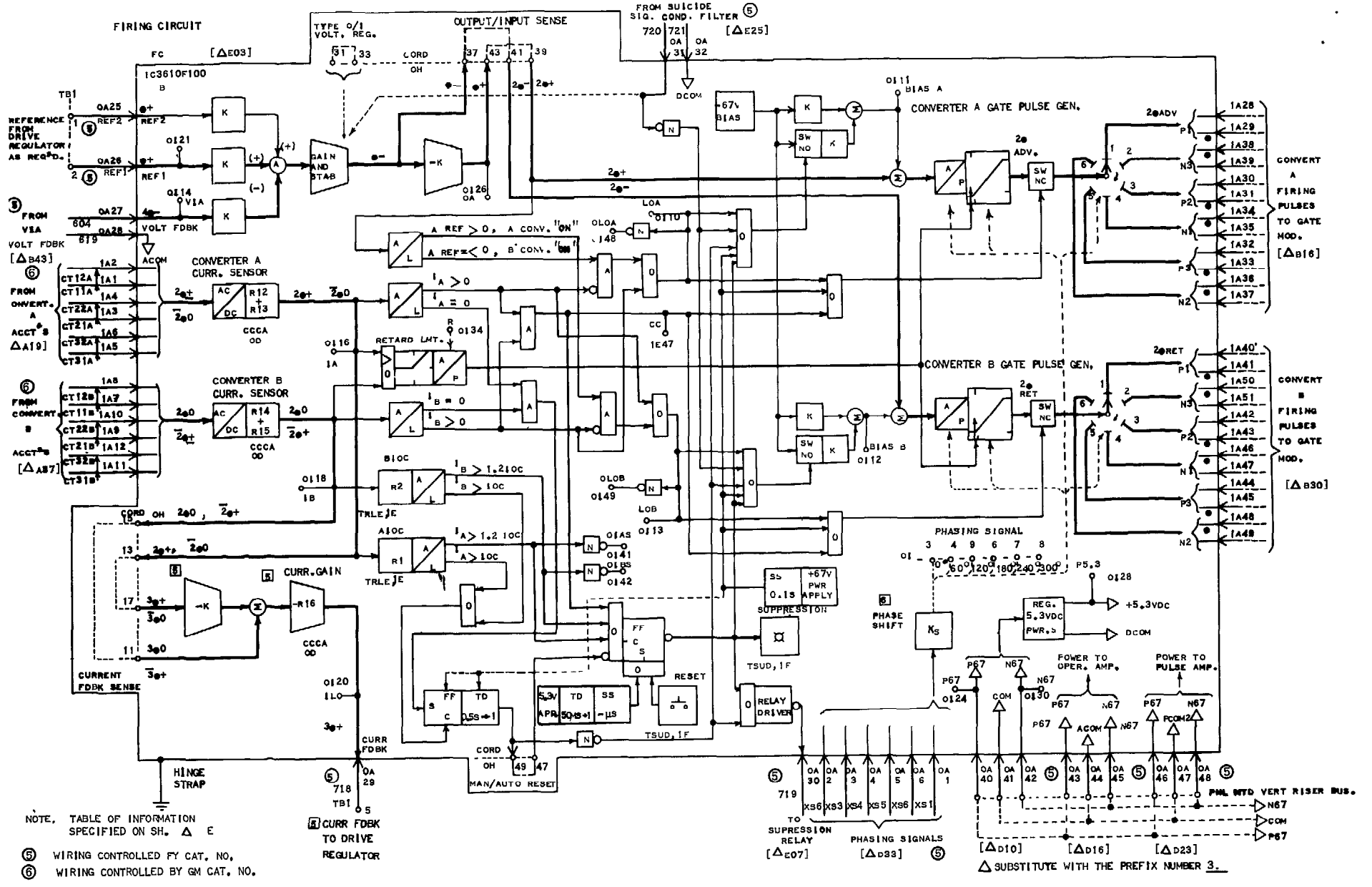
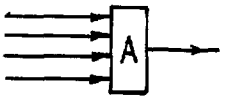
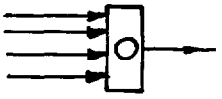
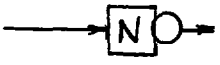

FIRING CIRCUIT
FC [△E03]
IC3610F100
B
TYPE 0/1 VOLT. REG.
OUTPUT/INPUT SENSE
FROM SUICIDE SIG. COND. FILTER ⑤
720 721 [△E25]
CORD OH
TB1
REFERENCE FROM DRIVE REGULATOR AS REQ'D.
REF2
REF1
VIA VOLT FDBK
[△B43]
FROM CONVERT. A ACCT'S △A19]
FROM CONVERT. B ACCT'S [△A37]
ACOM
CONVERTER A CURR. SENSOR
CONVERTER B CURR. SENSOR
CCCA OD
RETARD LMT.
TRLE
CURR. GAIN
CURRENT FDBK SENSE
HINGE STRAP
CURR FDBK
MAN/AUTO RESET
GAIN AND STAB
BIAS A
BIAS B
CONVERTER A GATE PULSE GEN.
CONVERTER B GATE PULSE GEN.
CONVERT A FIRING PULSES TO GATE MOD. [△B16]
CONVERT B FIRING PULSES TO GATE MOD. [△B30]
PHASING SIGNAL
PHASE SHIFT
SUPPRESSION
TSUD, 1F
RESET
RELAY DRIVER
REG. 5.3VDC PWR.S
+5.3VDC
DCOM
POWER TO OPER. AMP.
POWER TO PULSE AMP.
TO SUPRESSION RELAY [△E07]
PHASING SIGNALS [△D33]
PHL MTD VERT RISER BUS.
[△D10] [△D16] [△D23]
NOTE, TABLE OF INFORMATION SPECIFIED ON SH. △ E
⑤ WIRING CONTROLLED FY CAT. NO.
⑥ WIRING CONTROLLED BY GM CAT. NO.
△ SUBSTITUTE WITH THE PREFIX NUMBER 3.

Figure 4

Firing Circuit Description (cont'd)

In order to read the symbol diagram it is important that the reader understands how to interpret the basic logic and analog symbols used.

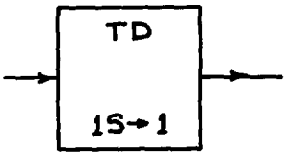
SYMBOL	DESCRIPTION
BASIC LOGIC ELEMENTS	
1. AND 	This symbol represents a logic "AND" element. The symbol, as shown, is to be interpreted as follows: "1's" are required on all four (4) inputs to obtain a "1" output. Conversely, a "0" on one or more inputs will result in a "0" output.
2. OR 	This symbol represents a logic "OR" element. The symbol as shown is to be interpreted as follows: A single "1" at any one of the four (4) inputs is all that is required for a "1" output. Of course, more than one "1" input will also result in a "1" output. Conversely, "0's" on all four (4) inputs will result in a "0" output.
3. NOT (Element)  OR 	This symbol represents a logic "NOT" element. In other words, this is a logic inversion element where the input and output are of opposite states. Although the two symbols perform identical functions, their true states are read differently. The first symbol reads "0" input gives a "1" output. The second reads "1" input gives a "0" output. The selection of which symbol to use is dependent upon which true state makes the diagram more readable.

SYMBOL

DESCRIPTION

BASIC LOGIC ELEMENTS

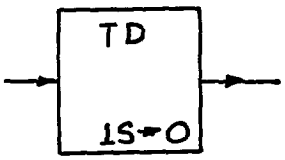
4. TIME DELAY



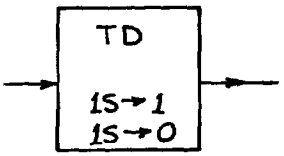
This symbol represents a logic "time delay" element. The first two (2) symbols, as shown, are to be interpreted as follows:

One (1) second after application of a "1" input will result in a "1" output, as indicated by the expression, 1S ⟶ 1.

Other Examples



The time delay may be in the direction of an output "0", as indicated by the expression, 1S ⟶ 0, denoting a one (1) second time delay to "0" output.



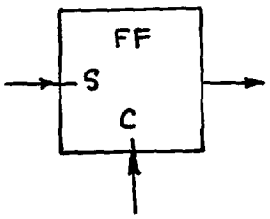
The following abbreviations of the units of time delay should be used:

us = Microsecond
ms = Millisecond
s = Seconds
m = Minutes
h = Hours

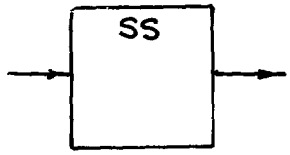
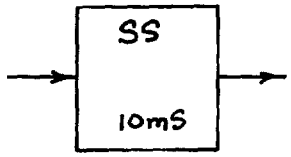
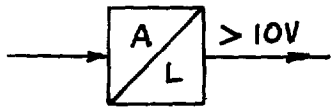
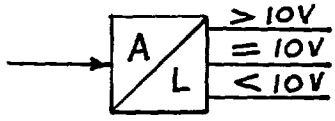
The numeric value and time unit is determined by its application.

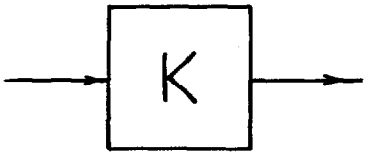
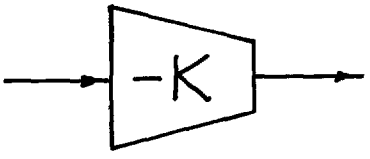
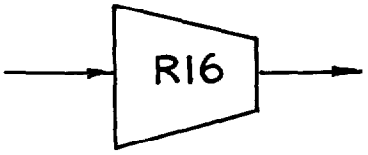
Special Purpose Logic Functions: These are frequently used logic functions which, as a general rule, can be constructed from basic logic elements.

5. FLIP-FLOP



This symbol represents a logic "flip-flop" which is a logic device that has memory or storage properties. By definition, it is a two (2) input device. The S (or "set") input inserts a signal which is stored on output. The C (or "clear") input removes the stored signal output. Furthermore, a "clear" signal input will be preferential over a "set" signal input. The symbol, as shown, is to be interpreted as follows:

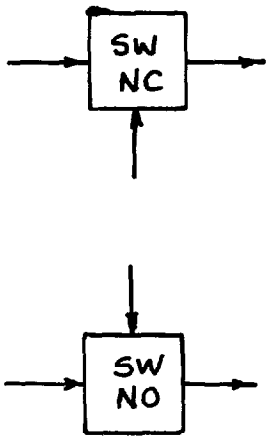
SYMBOL	DESCRIPTION
5. FLIP-FLOP - continued	A "1" on the S (or "set") input will cause a "1" on the output, which will remain "1" when the S input signal is returned to a "0". A "1" on the C (or "clear") input will cause the output to go to a "0", which will remain "0" when the C input signal is returned to a "0". Should a "1" appear simultaneously on both the S and C input terminals, then the output will remain a "0" since the C input always has preference over S input.
6. SINGLE SHOT  	This symbol represents a logic single shot, a logic device that emits a pulse output. The symbol, as shown, is to be interpreted as follows: A "1" input will cause a "1" output pulse. Changing the input to "0" will have no affect on the output. For applications where pulse width is important, its duration should be specified as shown. Use the same units of time for TIME DELAY logic function.
7. ANALOG TO LOGIC CONVERTER  	This represents a logic function that converts an analog input to a logic output in accordance with the expression shown above its output. The first symbol, as shown, is to be interpreted as follows: An analog signal input greater than 10 volts will cause a "1" output. The second symbol as shown is to be interpreted as follows:

SYMBOL	DESCRIPTION
7. ANALOG TO LOGIC CONVERTER (continued)	There will be three (3) logic outputs. One will be a "1" when input greater than 10V; another, a "1" when input equal to 10V; and another, a "1" when input less than 10V. The following signs should be used to express the relationship to a specified analog input: = Equals to <> Not equal to. < Less than. > Greater than. <= Less than or equal to. >= Greater than or equal to.
8. 	This represents a linear transfer function K in which there is no inversion between input and output. Examples: resistors, resistance bridges, non-inverting amplifiers, etc.
9.  	Same as above except that there is an inversion between input and output. Where it is desired to indicate a particular component that affects the gain of a transfer function this symbol is used. It indicates that the gain of the transfer function can be changed by adjustment of R16, not that the gain is R16.

SYMBOL

DESCRIPTION

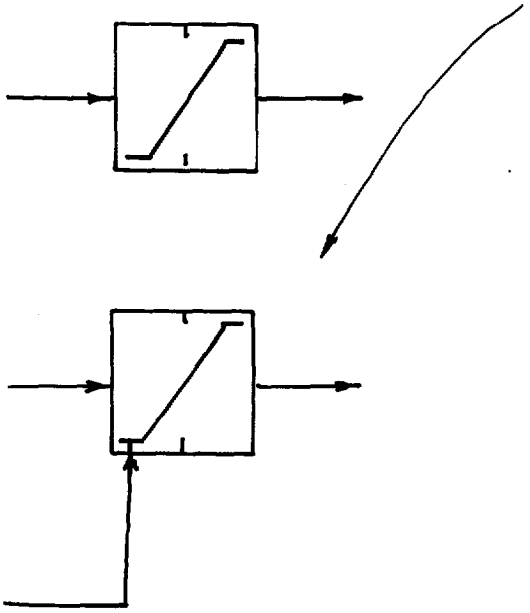
10.



This represents a switch that gates the output signal with respect to its input signal. The input is on the left and the output is on the right. The signal for operating the switch may be shown along the upper or lower side of the box. The letters NC imply that the switch would be closed if the switch operating signal were removed. The letters NO imply that the switch is open under the same conditions.

Examples: interlocks, SCR's, transistors, etc.

11.



Represents a linear transfer function in which the output has positive and negative limits.

This example shows that the negative limit is controlled by another signal.

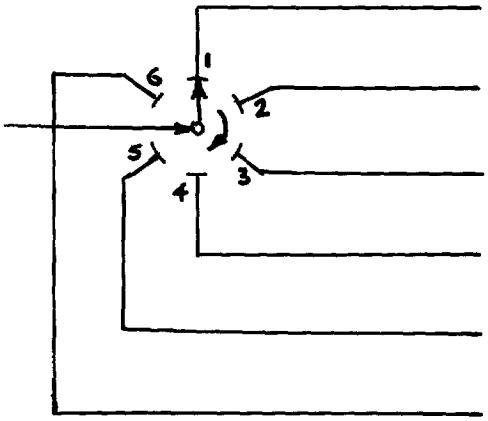
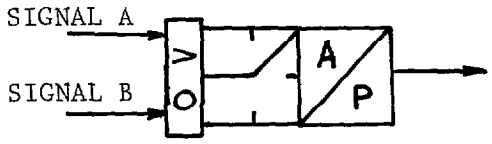
SYMBOL	DESCRIPTION
12. 	This symbol represents the generation of 6 pulses which appear sequentially 1 thru 6.
13. 	This symbol shows the way that various symbols can be combined. It is to be interpreted in the following way: The greater of the two analog signals (signal A and signal B), if positive produces a control signal in the pulse domain.

Fig. 4, Sheet 2-3 showed the complete firing circuit symbol in detail. The following is a breakdown of the firing circuit describing in detail each portion of the symbol:

INTERNAL VOLTAGE REGULATOR:

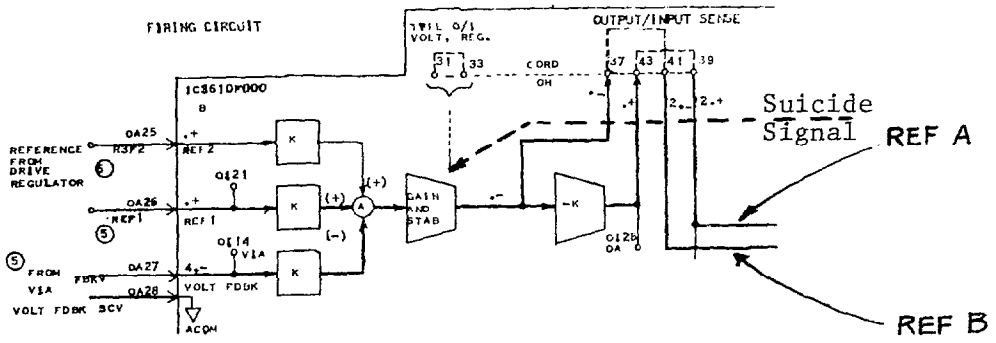


Figure 5

The above portion of the firing circuit symbol shows the generation of the two analog signals, Ref. A and Ref. B, which are applied to the A&B gate pulse generators.

All reversing and some non-reversing Siltrol drives have an internal voltage regulator. The transfer function for the Siltrol for drives with the internal voltage regulator is:

$$\frac{Ed}{15\ [1000]}$$

The first operational amplifier provides the necessary gain and stabilization circuitry and is connected as shown on Figure 6.

The "suicide signal" shows a relationship exists between the suicide signal and the gain and stabilization amplifier. The relationship is that whenever a process regulator suicide signal is present a 100Ω resistor is connected around the amplifier.

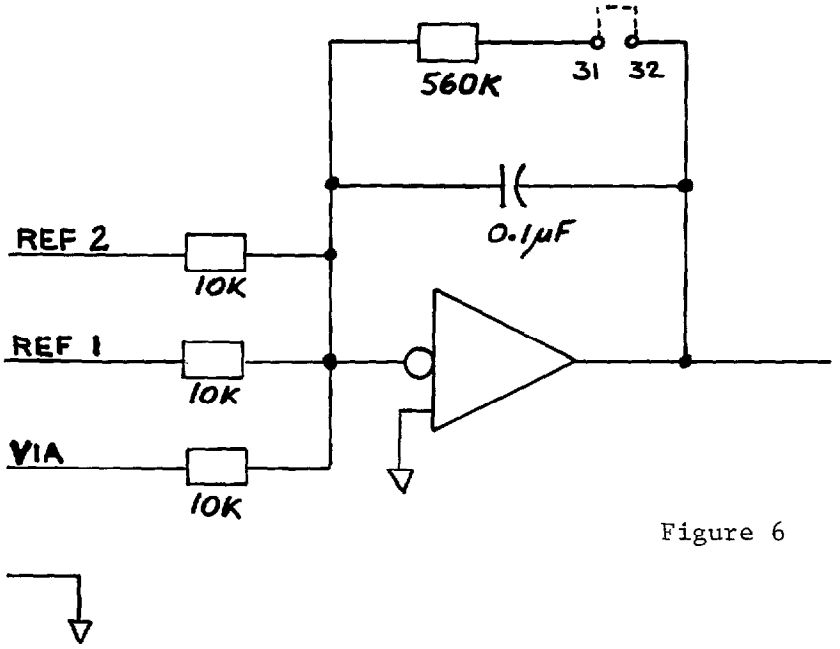


Figure 6

For a type 0 system pins 31 and 32 of the coordination card are jumpered to give the amplifier a finite gain of 56 and a downbreak at approximately 20 radians.

$$\frac{56}{[20]}$$

For a type 1 system, the jumper is not connected and the amplifier has a gain of:

$$\frac{1000}{p}$$

The second operation amplifier has unity gain and simply inverts the output of the first amplifier so that both negative and positive signals (Ref. A & Ref. B) are available to drive A & B gate pulse generators. On a non-reversing drive this amplifier is obviously not required and therefore not provided.

Note on Figure 5 that by connections on the coordination card CORD (Slot 0H), the output/input sense of the Siltrol can be inverted. IF FOR ANY REASON THE INPUT/OUTPUT SENSE OF A SILTROL IS CHANGED THEN THE VOLTAGE FEEDBACK SENSE MUST BE CHANGED BY INTERCHANGING THE CONNECTIONS IN1 & IN2 ON THE VOLTAGE ISOLATION AMPLIFIER VIA.

On some non-reversing applications an internal voltage regulator is not required. In which case, the operational amplifier has a 10K feedback resistor to give unity gain.

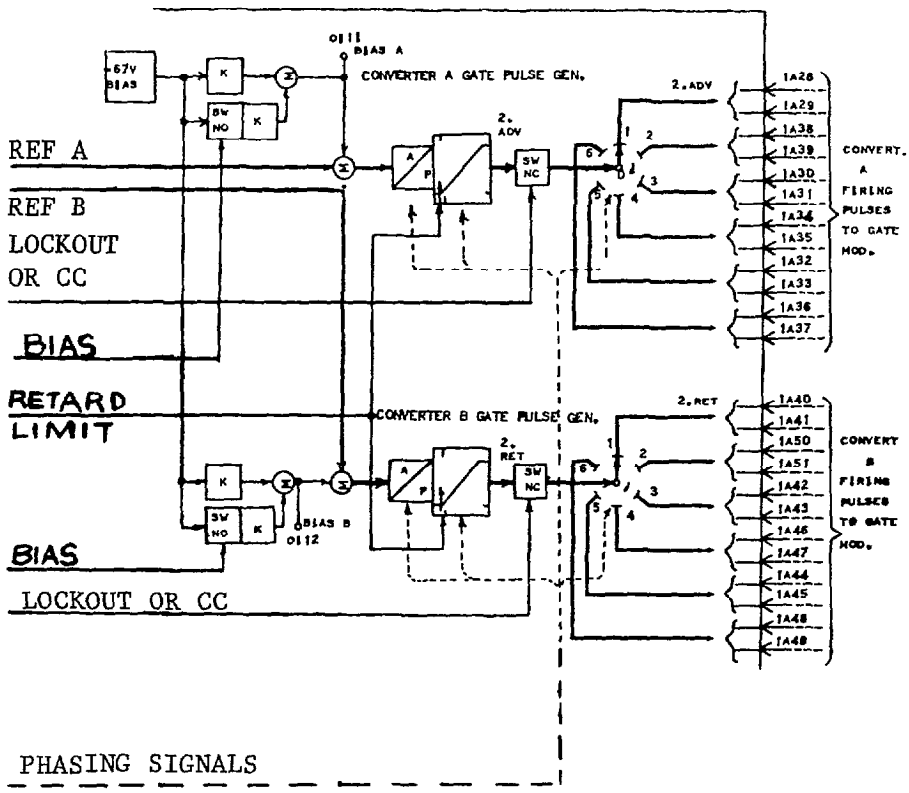


Figure 7

Reference A and Reference B are applied to converters A and B Gate Pulse Generators respectively.

Considering Reference A only, it is summed with Bias A and with 6 sine wave phasing signals to produce 6 firing pulses at the desired phase angle. (The operation of the gate pulse generator is discussed in detail later.)

Similarly, Reference B is summed with Bias B and 6 sine wave phasing signals. (The functions of the signals: Bias Lockout, Circulating Current and Retard Limit signals are discussed in detail later.)

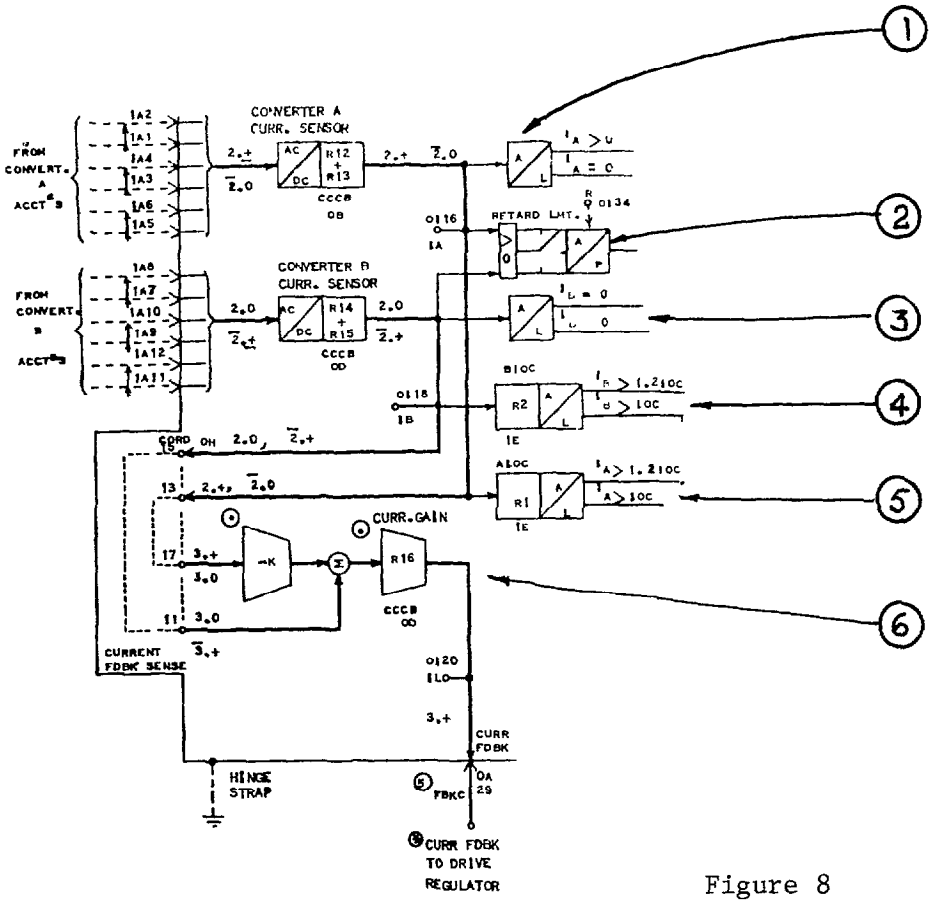


Figure 8

The a-c lines, L1-L2-L3, which feed the A & B thyristor bridges are each fed through a current transformer.

There are 6 CT's employed; 3 for the A bridge and 3 for the B bridge.

The outputs of the CT's are a-c and are rectified in the "AC/DC" "current sensor", as shown in the Figure 8. The nature of the CT is such that a burden resistor is required to attain a specific voltage at a specific a-c line current. These resistors (R12, R13, R14, and R15) are, therefore, selected on a per drive basis to obtain the desired output voltage for rated current.

The d-c output voltages of the A and B current sensors are designated I_A and I_B respectively. These signals are used in the inverter limit circuitry, the reversing logic circuitry and the operational amplifiers to generate the process regulator current feedback (I_L).

Current Feedback (continued) -

To provide the necessary logic signals for the inverter limit and reversing logic circuitry various analog to logic signals are generated as shown in Figure 8 by symbols (1) through (5).

(1) and (3) - These symbols show the generation of two logic signals each from the I_A and I_B analog signals.

$I_A > 0$ - A logic "1" is produced when I_A is greater than 0.

$I_A = 0$ - A logic "1" is produced when I_A is equal to 0.

$I_B > 0$ - A logic "1" is produced when I_B is greater than 0

$I_B = 0$ - A logic "1" is produced when I_B is equal to 0.

(2) - This symbol shows the production of a signal to program the retard limit as a function of I_A or I_B.

(4) and (5) - These symbols show the generation of logic signals produced by overcurrent conditions.

$I_A >$ IOC - This is a logic "1" when I_A is greater than 1 x the overcurrent setting, the level of which is adjusted by R1. IOC level is 1.1 times the maximum rating (e.g. 300% for 10 sec.); IOC level = 1.1 x 300%.

$I_A > 1.2$IOC - This is a logic "1" when I_A is greater than 1.2 x the overcurrent setting; 1.2 x 1.1 x 300%. This level is automatically obtained by adjustment of R1 for the 1.0 overcurrent setting. There is no independent adjustment.

Similarly for package B, logic signal "1's" are obtained for $I_B >$ IOC and $I_B > 1.2$ IOC with adjustment by R2.

(6) - The outputs of A & B current sensors are positive voltages designated I_A and I_B and are used to generate I_L (the drive regulator current feedback signal) via two operational amplifiers. As connected on Figure 8 I_A is fed through a unity gain inverting amplifier and its output ($-I_A$) is summed with I_B. Since I_A and I_B are such that both cannot be present simultaneously, either $-I_A$ or I_B will drive the second amplifier. The gain of the second amplifier is adjustable by R16 on card CCCA slot OD to give the desired output gradient.

With the coordination card connections as shown on Figure 8 and with current flowing in package A, I_L will be a positive voltage. With current flowing in package B, I_L will be a negative voltage. This sense can be inverted by changing the pin connections 15, 13, 17 and 11 on the coordination card.

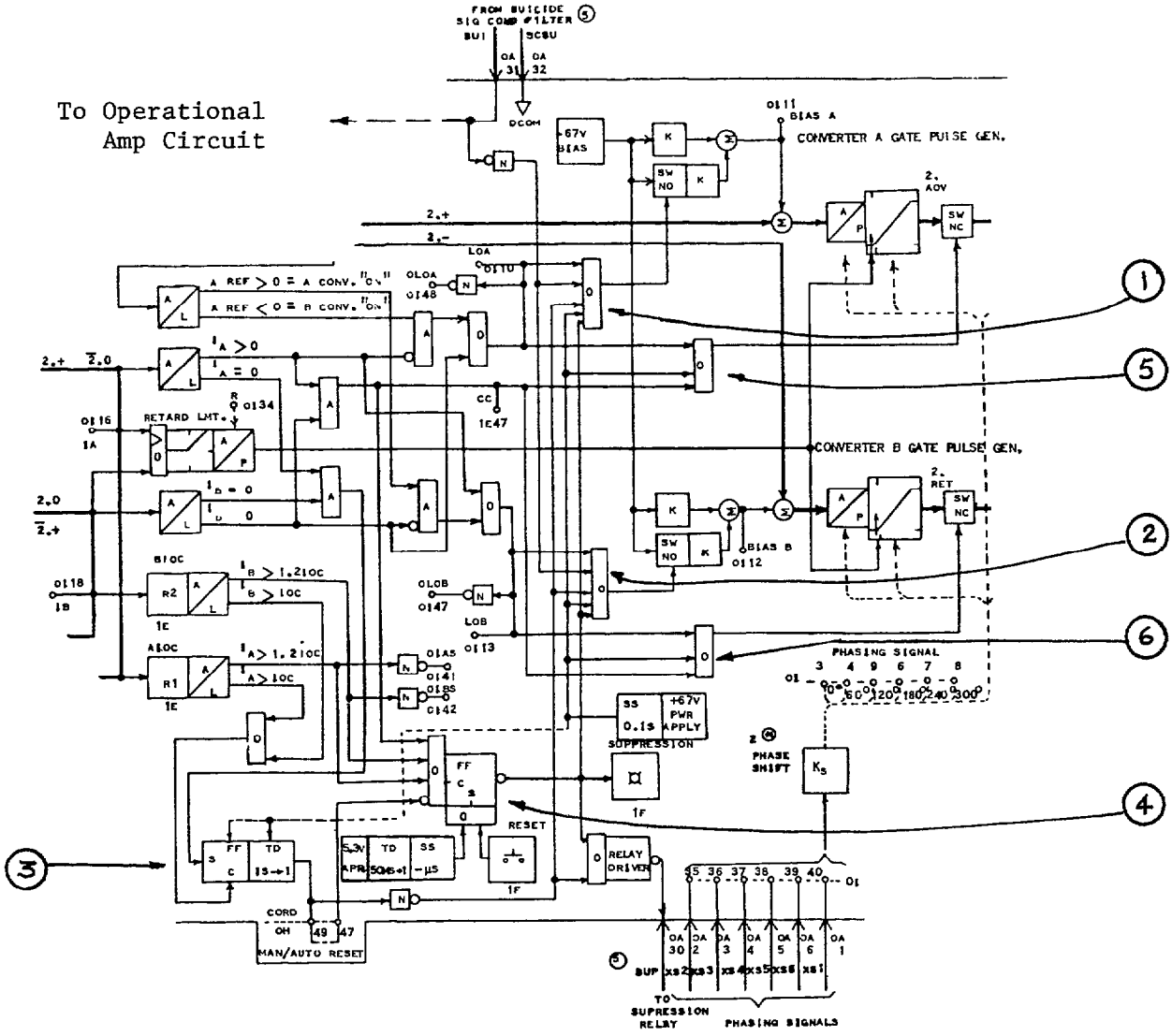


Figure 9

This portion of the firing circuit shows the various logic functions performed to provide the necessary protective features.

(1) and (2) - The outputs of these "OR" gates initiate a strong negative bias signal (-48 volts) to the A or B gate pulse generators, retarding pulses to retard limit.

REVERSING LOGIC AND OVERCURRENT PROTECTION (Cont'd)

The various inputs to the "OR" gates are as follows:

LOA (Lock out A) - This is a logic signal applied to (1) and generated as a function of Reference A, I_A, and I_B.

LOA is logic "1" if $I_B > 0$ OR if (Ref. A $<= 0$ and I_A is not > 0).

The second input to (1) is a suicide signal from the drive regulator retarding pulses to retard limit when the drive regulator is suicided. This signal is also applied to (2).

The third input to (1) is generated from the overcurrent logic which is discussed later. This signal is also applied to (2).

The fourth input to (1) is a 0.1 second pulse retarding pulses during application of 67V power. This signal is also applied to (2).

The fifth input to (1) is also generated from the overcurrent protection which is discussed later. Again, this signal is also applied to (2).

LOB (Lock Out B) - This is a logic signal applied to (2) and generated as a function of Ref. A, I_A and I_B.

LOB is logic "1" if $I_A > 0$ OR (Ref. A > 0 and I_B is not > 0).

(3) - This flip-flop is "set" on application of 67V power or when -

($I_A = 0$ and $I_B = 0$)

When an IOC (first level) overcurrent condition exists, the flip-flop is "cleared" and gives a logic "0" output. The clear signal is obtained when -

($I_A >$ IOC or $I_B >$ IOC)

The "0" output is inverted and applied to OR gates (1) and (2) to bias both A and B gate pulse generators to retard limit. It is also applied to a relay driver to operate the suppression relay and trip out the drive.

The flip-flop has a time delay of 0.5 second when going from "0" to "1" to allow time for the drive to trip out and prevent mal-function by the current going to zero and resetting the flip-flop before the drive contactor has opened.

(4) - This flip-flop applies a strong bias via OR gates (1) and (2) and trips out the drive via the suppression relay for fault conditions that must be manually reset.

The flip-flop is "set" by application of 5.3V power OR by the RESET pushbutton.

The flip-flop is "cleared" when (I_A > 0 and I_B > 0) circulating current fault.

OR

(I_A > 1.2 IOC)

OR

(I_B > 1.2 IOC)

} Second Level Overcurrent

OR

(I_A > IOC)

OR

(I_B > IOC)

} First level overcurrent if connected via the coordination card pins 47 and 49.

(5) - This OR gate shows that in addition to retarding pulses to retard limit, gate pulses are suppressed for the following conditions:

LOA - Lock out A

CC - Circulating Current (current in packages A & B)

67V Power Apply - Pulses are inhibited for 0.1 seconds during application of power.

(6) - Similarly, for B gate pulse generator, pulses are suppressed for the following conditions:

LOB - Lock out B
CC - Circulating Current
67V Power Apply

GATE PULSE GENERATOR:

Throughout the gate pulse generator description, reference is made to phasing signals at various phase angles. Refer to Figure 10 for definition of these angles. The figure shows the relationship between the phasing signals and the power bridge inputs. (Power transformer secondary).

0° is defined as the power bridge input Line 1 to neutral.

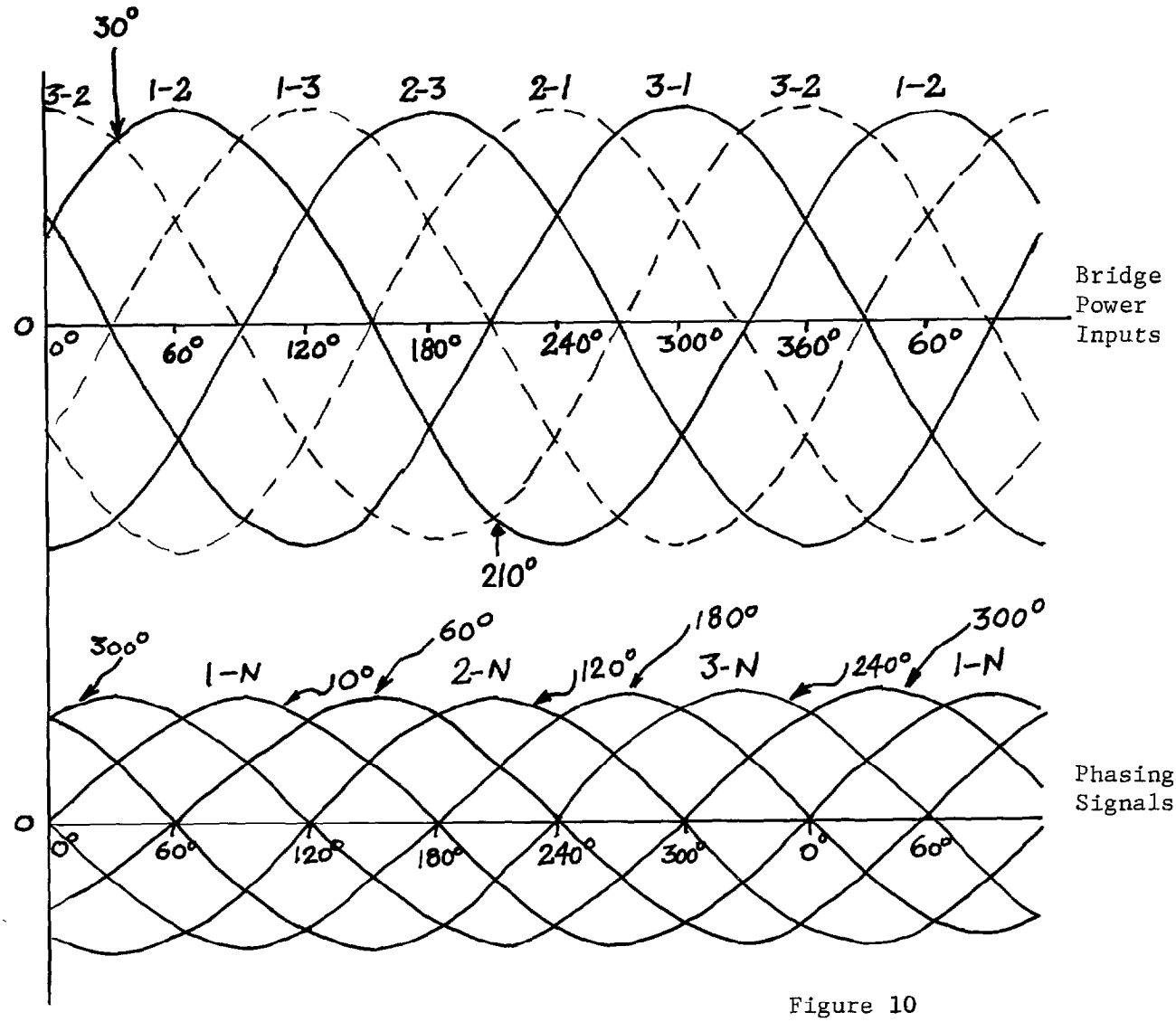


Figure 10

GATE PULSE GENERATOR (continued)

The phasing signals are derived from a phasing transformer which is a 50/60 cycle, three phase, 250VA device. The primary is normally connected in delta and the secondary in a six phase star. These six star windings are phasing reference waves. They are fed into a phase shift network card where they are conditioned for use in the firing circuit by means of RC filters. If the power transformer for the Siltrol is delta-delta connected, a thirty degree phase shift is introduced to the reference in the phase shift network. If the power transformer is delta wye connected, a zero shift is required in the phase shift network. However, the reference still requires filtering to avoid dips and spikes which are sometimes present on the a-c bus. Therefore, the filtering is designed to introduce a 60° phase shift. Because the 6 reference waves are 60° apart, the output of the phase shift network is connected to give the desired zero phase shift.

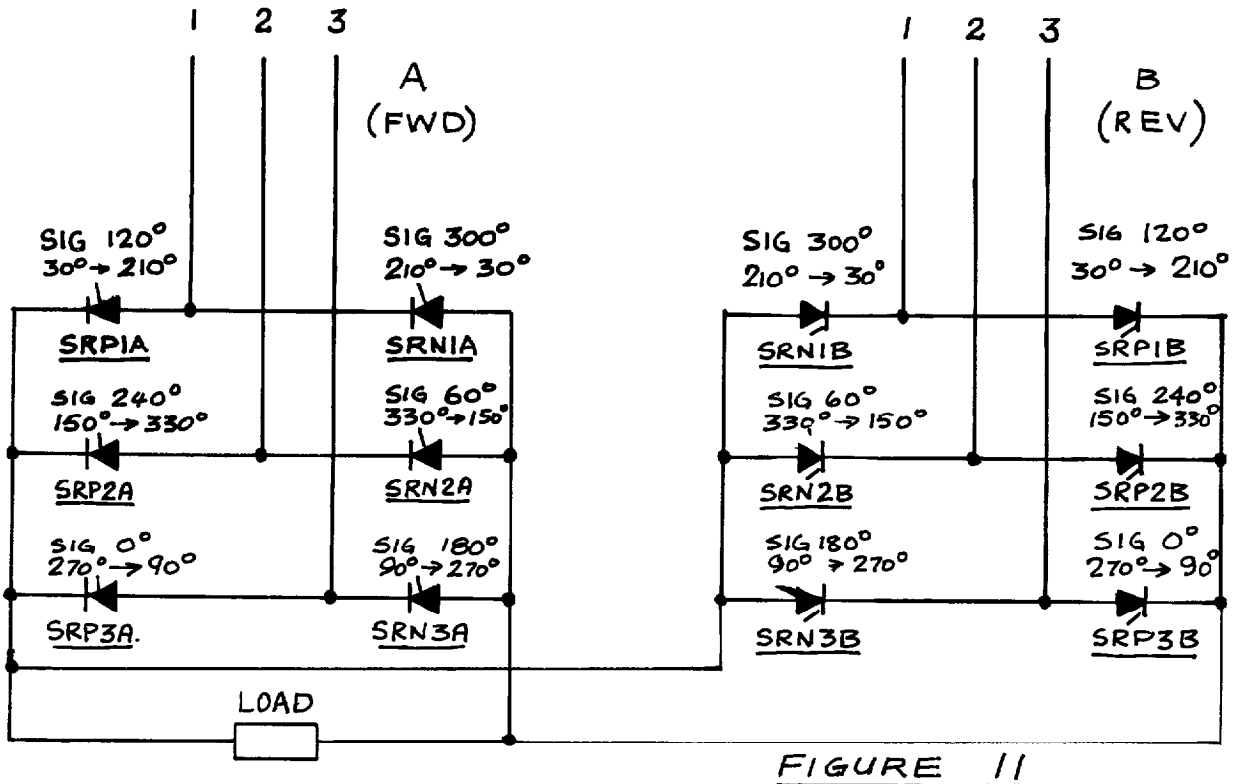


FIGURE 11

Consider the firing of cell SRP1A. Referring to figures 10 and 11, the allowable firing period for cell SRP1A is from 30° ⟶ 210°. Actually, to ensure good commutation it is desirable to limit full advance to around 36°. It is also necessary to limit maximum retard to around 185° but to advance the retard limit as a function of current, to ensure good commutation when inverting.

To fully understand how the advance and retard limit function, we will first consider the operation of the gate pulse generator card.

Gate Pulse Generator (Continued) -

Figure 12 is a block diagram showing functionally how a gate pulse generator card operates. There are 2 cards, one for each power bridge, A & B.

The various signal inputs to the card are as follows:

REF - For power bridge A, this is reference A, the output of the regulating amplifier, a DC voltage variable between ±25 volts.

BIAS - For power bridge A, this is Bias A, a DC level generated by the suppression and Bias card. It has two voltage states: -10V and -48V.

SIGNAL - This is the name assigned to the 6 sine wave references.

INVERTER LMT - These inputs are the 6 inverter limit inputs required to control the maximum retard angle and are discussed in detail later.

LOCKOUT - This input is derived from the reversing logic; its function having already been discussed.

The symbol for pulse generation for one cell only is shown in detail. The circuits for the other five are identical as indicated.

Considering P1 (cell SRP1A), it can be seen that "BIAS" and "REF" are summed with the 120° sine wave and the resultant if > 0 gives a logic "1". This is then applied to an OR gate with the inverter limit input. The output of the OR gate is applied to an AND gate with the advance limit signal (ADV LMT).

The output of the AND gate initiates a 50 μsec "0" pulse on P and "sets" a flip-flop, giving a "0" output on OGP. The flip-flop is "cleared" when the flip-flop of circuit P2 is "set" (120° later). Similarly, P2 is reset by P3 and P3 is reset by P1. The flip-flop is also reset whenever a lockout condition exists.

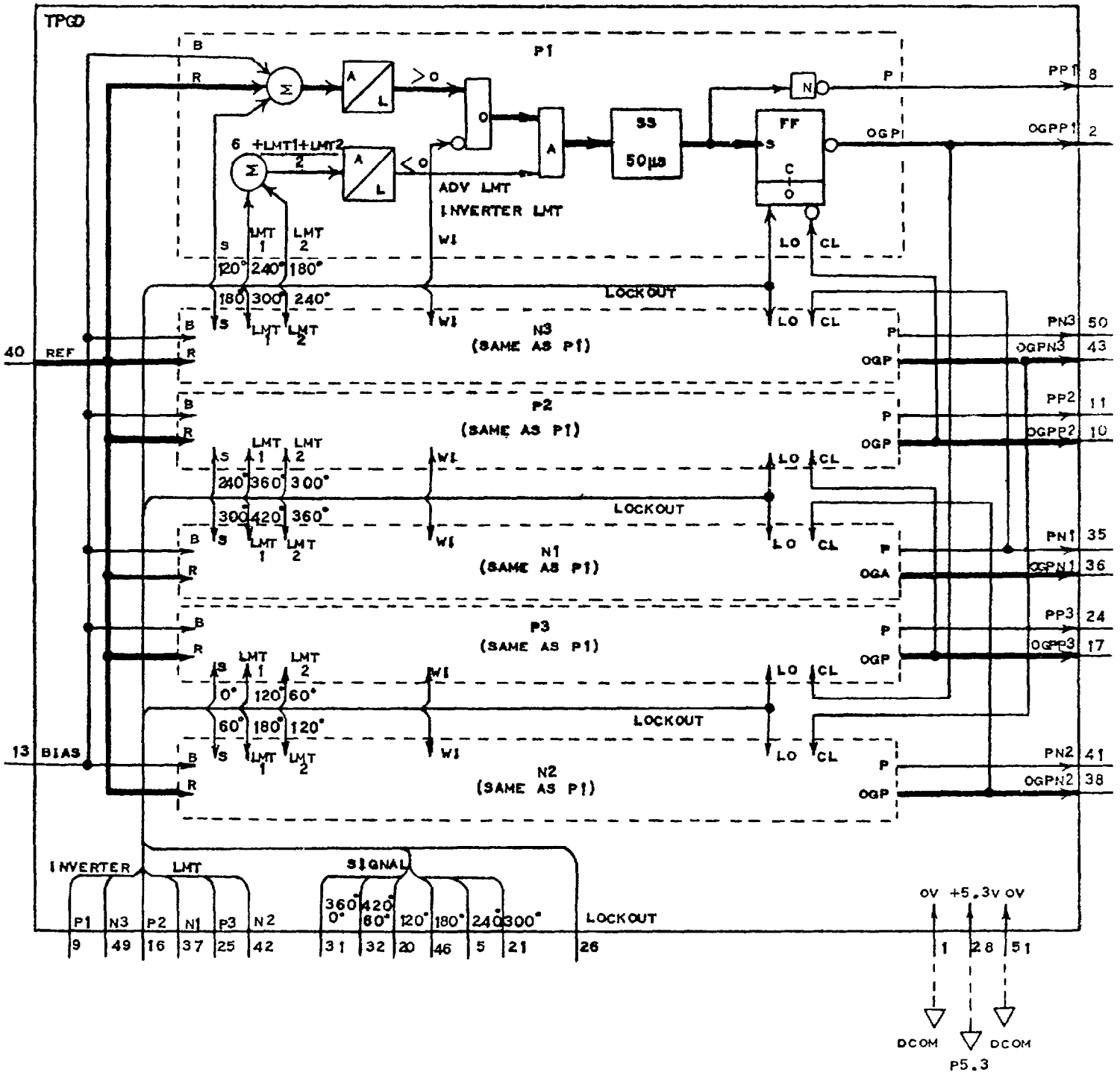
Figure 13 shows the generation of P and OGP for various values of REF.

The point at which pulses are generated during the allowed firing period (between advance limit and inverter limit) is determined by BIAS, REF, and for cell P1, the 120° sine wave. If BIAS is -10V and REF is +10V, then the cell will fire just as the 120° wave goes from a negative value to a positive value. On an inductive load, this will give 0V output.

If BIAS is -10V and REF 0V, pulses will be generated at approximately 137° retard, resulting in discontinuous current.

With -10V BIAS and 25V REF, a pulse will be generated at full advance or approximately 30°.

Note that except when BIAS and REF are equal and opposite pulses automatically tend to advance for reduced line voltage and retard for increased line voltage.



NOTE : ALL DEGREES SHOWN ARE UNDERSTOOD TO BE DEGREES LAGGING.

Figure 12

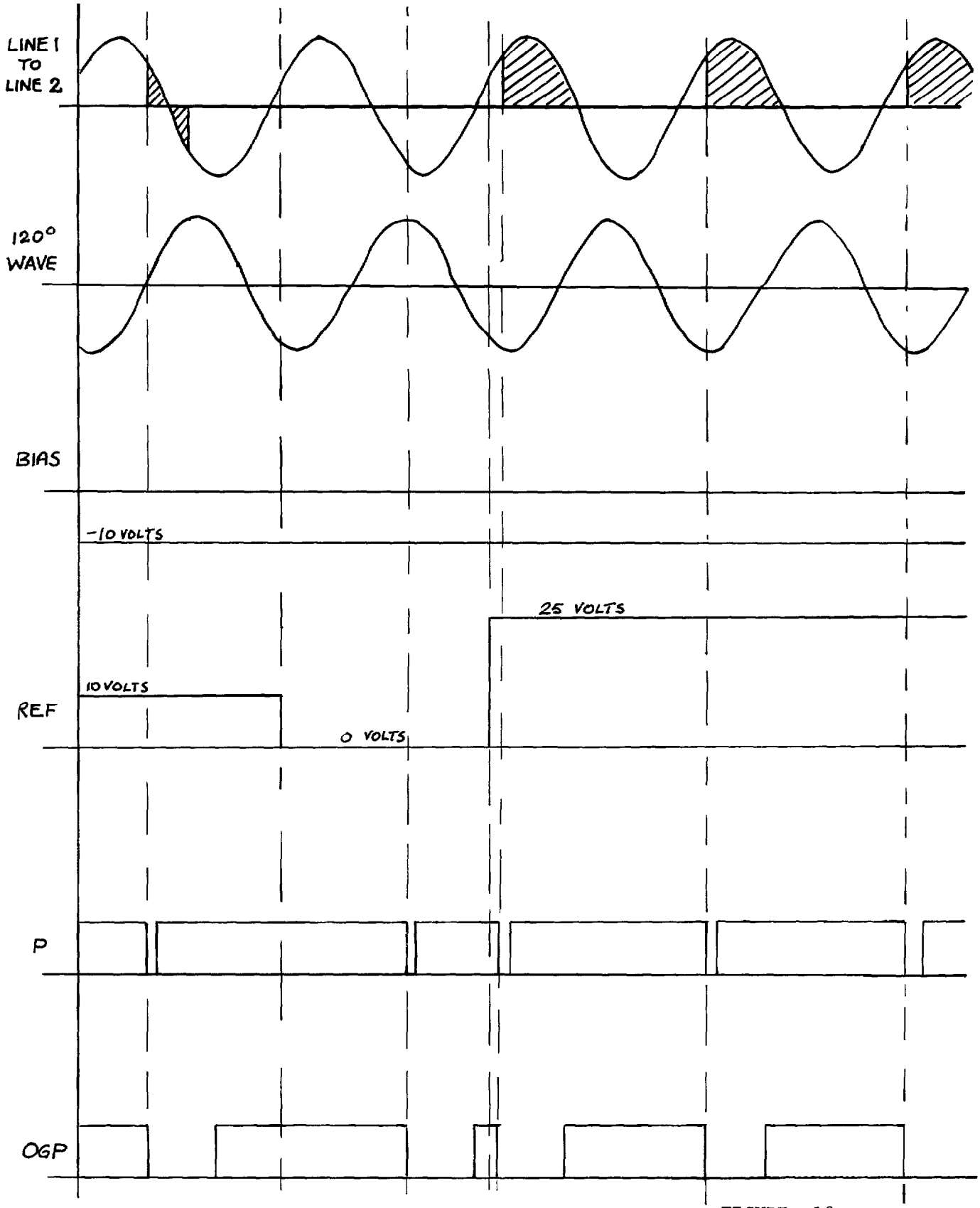

LINE 1
TO
LINE 2
120°
WAVE
BIAS
-10 VOLTS
25 VOLTS
10 VOLTS
REF
0 VOLTS
P
OGP

FIGURE 13

ADVANCE LIMIT

The advance limit is required to inhibit full advance to 36°. Referring to Figure 12, it can be seen that it is obtained by correctly weighting the 180°, and 240° waves and adding them, producing a 216° wave (see Figure 14). A logic "1" is produced whenever the 216° wave is negative (< 0); i.e., a logic "1" is produced for the period 36° → 216° and, hence, inhibiting pulse generation at any other phase angles.

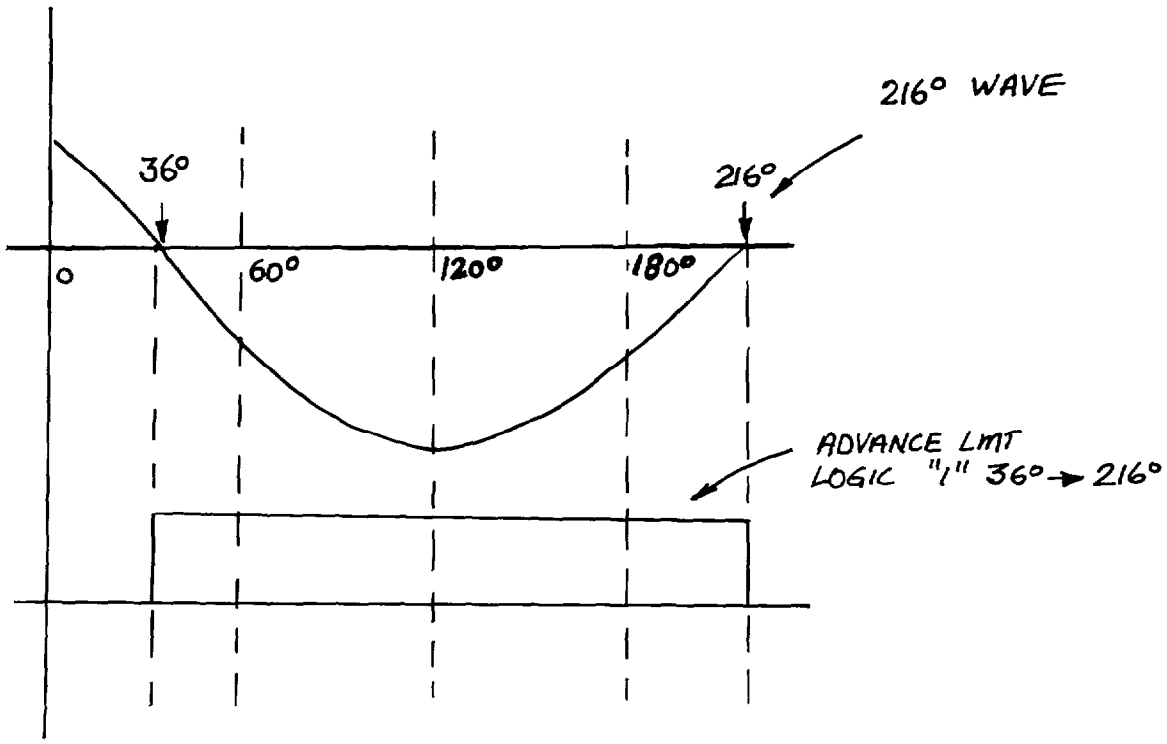


FIGURE 14

INVERTER LIMIT

Figure 15 is a symbol diagram showing functionally how the inverter limit card works. The symbol for P1 only is shown in detail.

The various inputs are as follows:

I_A & I_B - The current feedback signals are OR's as shown in Figure 15.

The OR symbol shows that if I_A or I_B are positive, then the output of the OR gate is an analog signal equal to the larger of I_A and I_B . Normally, I_A and I_B are not present simultaneously.

S1, S2 and S3 - These are the various sine waves required to produce the inverter limit function.

LOA, LOB - not used for Siltrol.

Figure 16 shows the various sine wave inputs and the resulting logic outputs. The 210° wave is obtained from the summation of the 180° and 240° waves. The 210° wave is then summed with the negative half of the 120° to give the third wave which, without Bias and I_A and I_B , inputs would be negative from approximately 330° to 210°. Referring to Figures 11 and 12, it can be seen that P1A and P1B are logic "0" from 210° to 330°.

The dotted lines "BIAS" and "I_A OR I_B " show how the retard limit angle advances when these d-c levels are introduced.

The dotted version of the third wave shows how the retard limit advances as a function of reduced line voltage.

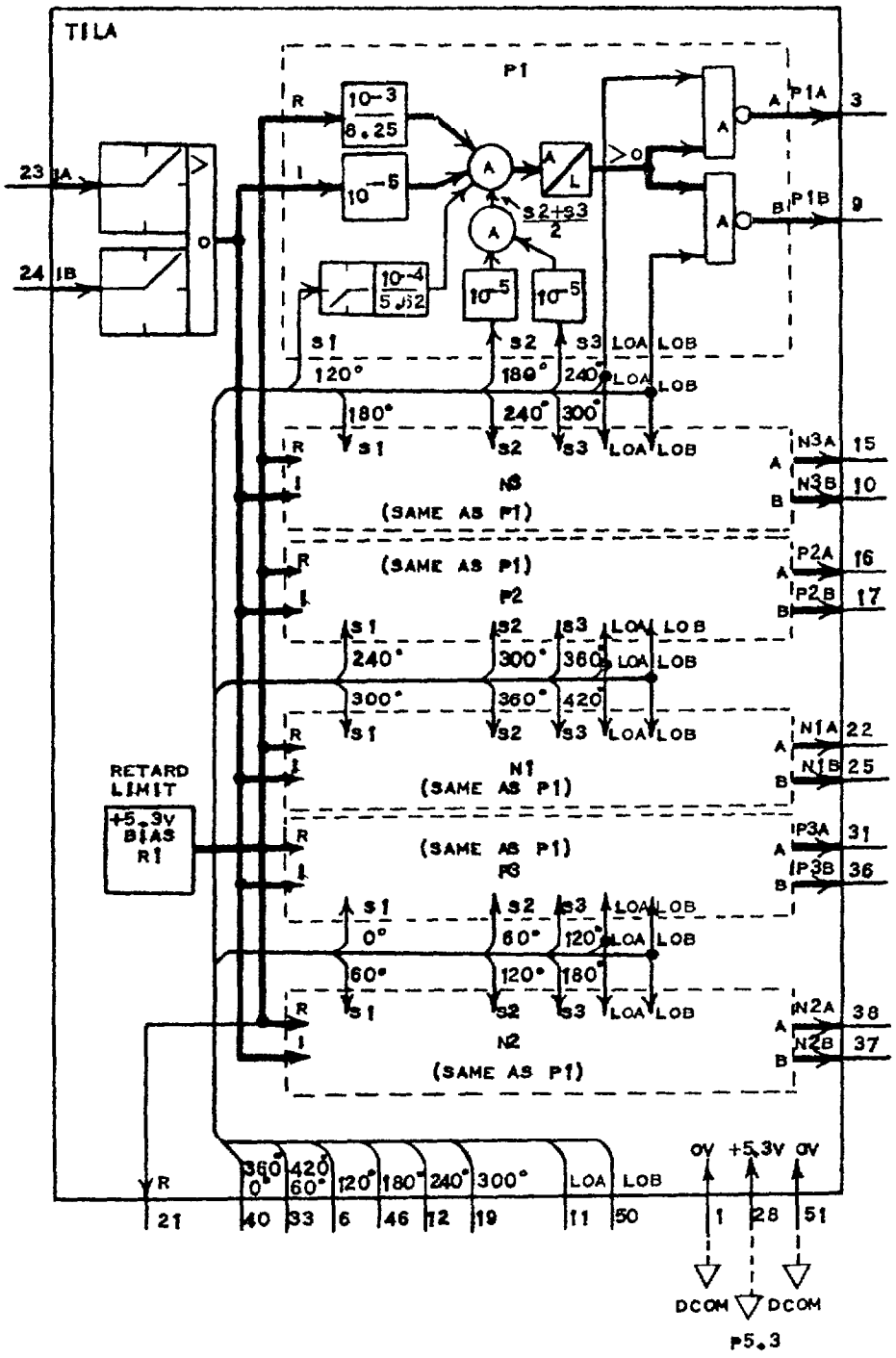

T1LA
P1
R
10⁻³ / 6.25
10⁻⁵
A
A/L
>0
S2+S3 / 2
10⁻⁴ / 5.62
10⁻⁵
10⁻⁵
23 1A
24 1B
O
S1
S2
S3
LOA
LOB
120°
180°
240°
300°
360°
420°
P1A 3
P1B 9
N3
(SAME AS P1)
N3A 15
N3B 10
P2
P2A 16
P2B 17
N1
N1A 22
N1B 25
P3
IP3A 31
IP3B 36
N2
N2A 38
N2B 37
RETARD LIMIT
+5.3V BIAS R1
0°
60°
LOA LOB
21 40 33 6 46 12 19 11 50
OV +5.3V OV
1 28 51
DCOM DCOM
P5.3

Figure 15

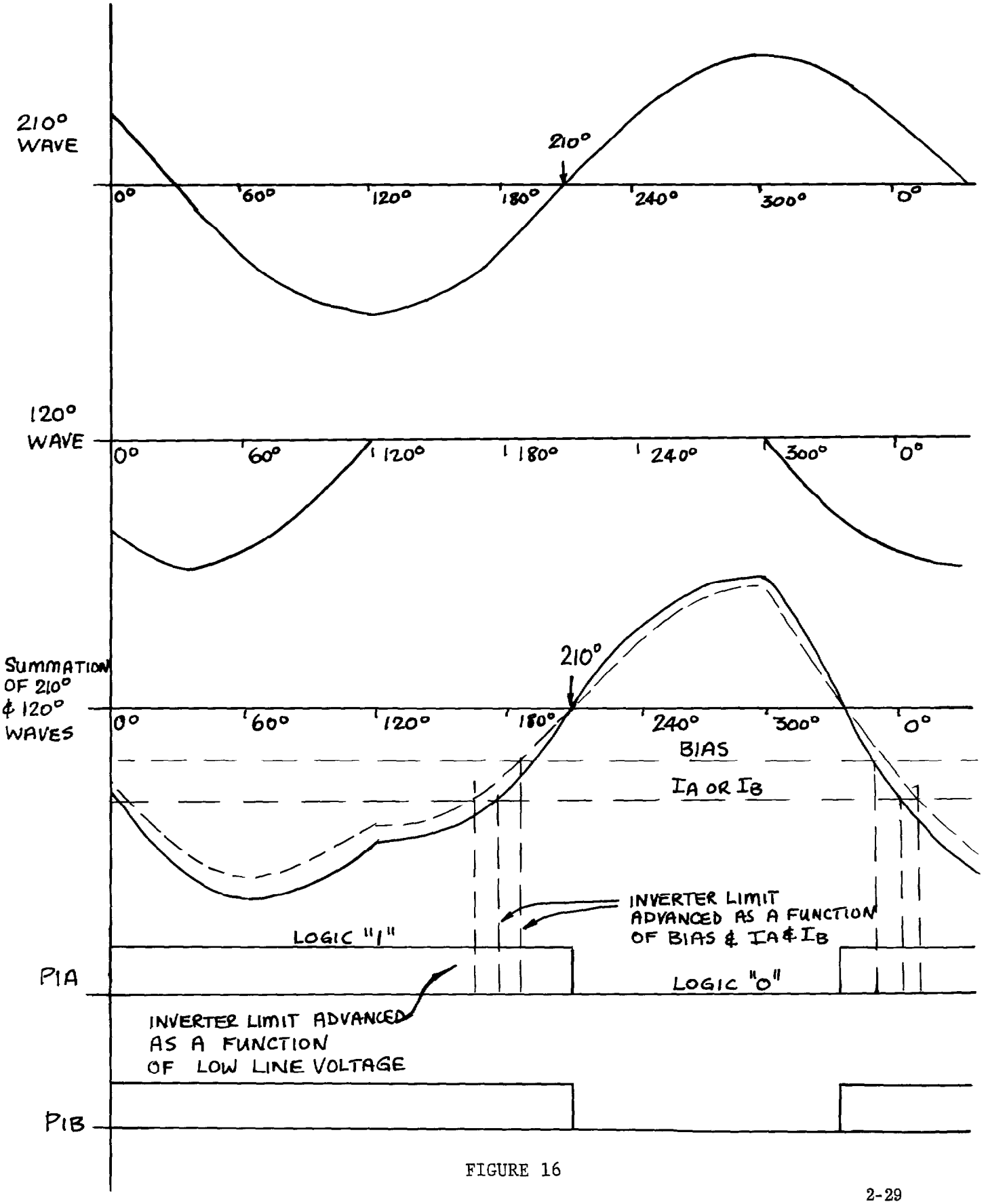

210° WAVE
210°
0°
60°
120°
180°
240°
300°
0°
120° WAVE
0°
60°
120°
180°
240°
300°
0°
SUMMATION OF 210° & 120° WAVES
210°
0°
60°
120°
180°
240°
300°
0°
BIAS
IA OR IB
INVERTER LIMIT ADVANCED AS A FUNCTION OF BIAS & IA & IB
LOGIC "1"
LOGIC "0"
P1A
INVERTER LIMIT ADVANCED AS A FUNCTION OF LOW LINE VOLTAGE
P1B

FIGURE 16

GATE PULSE AMPLIFIER

The gate pulse amplifier is designed to convert the low power microelectronic signal outputs of the gate pulse generator to relatively high power pulse trains, which are fed to pulse transformers (gating module). The pulse transformer outputs drive into the gates of the thyristors.

Figure 17 is a symbol showing functionally how the gate pulse amplifier operates.

The output OGP of the gate pulse generator is connected to terminal 2 (GLP1) of the gate pulse amplifier inverted in an OR gate (terminal 3 is not used) and then "ANDED" with the asymmetrical multi-vibrator output on P1 Pulse Amplifier. This gates the multi-vibrator output to the P1 Pulse Amp for the correct 120° period.

The output P of the gate pulse generator is connected to terminal 47 (PP1) of the gate pulse amplifier, and gated with the lockout signal in various ways as can be seen in Figure 17.

Assume that a lockout condition is present, in which case, there will be a "0" signal on the "FORCE 0" input to the asymmetrical multi-vibrator. This signal will force a "0" on the output of the vibrator inhibiting pulses.

If the lockout signal is removed, the vibrator starts oscillating ("0" for 40/μsec.; "1" for 20/μsec).

When a 50/μsec. pulse appears on PP1 (terminal 47) through the logic, a "FORCE 1" signal is generated, which forces a "1" on the output of the multi-vibrator, ensuring that the first pulse is of 50/μsec. duration, to ensure good firing.

The input on terminal 2 gates the pulse train through P1 Pulse Amplifier for the correct 120° period.

TP1 on pin 12 is a test point which reads the multi-vibrator output.

SYMBOL

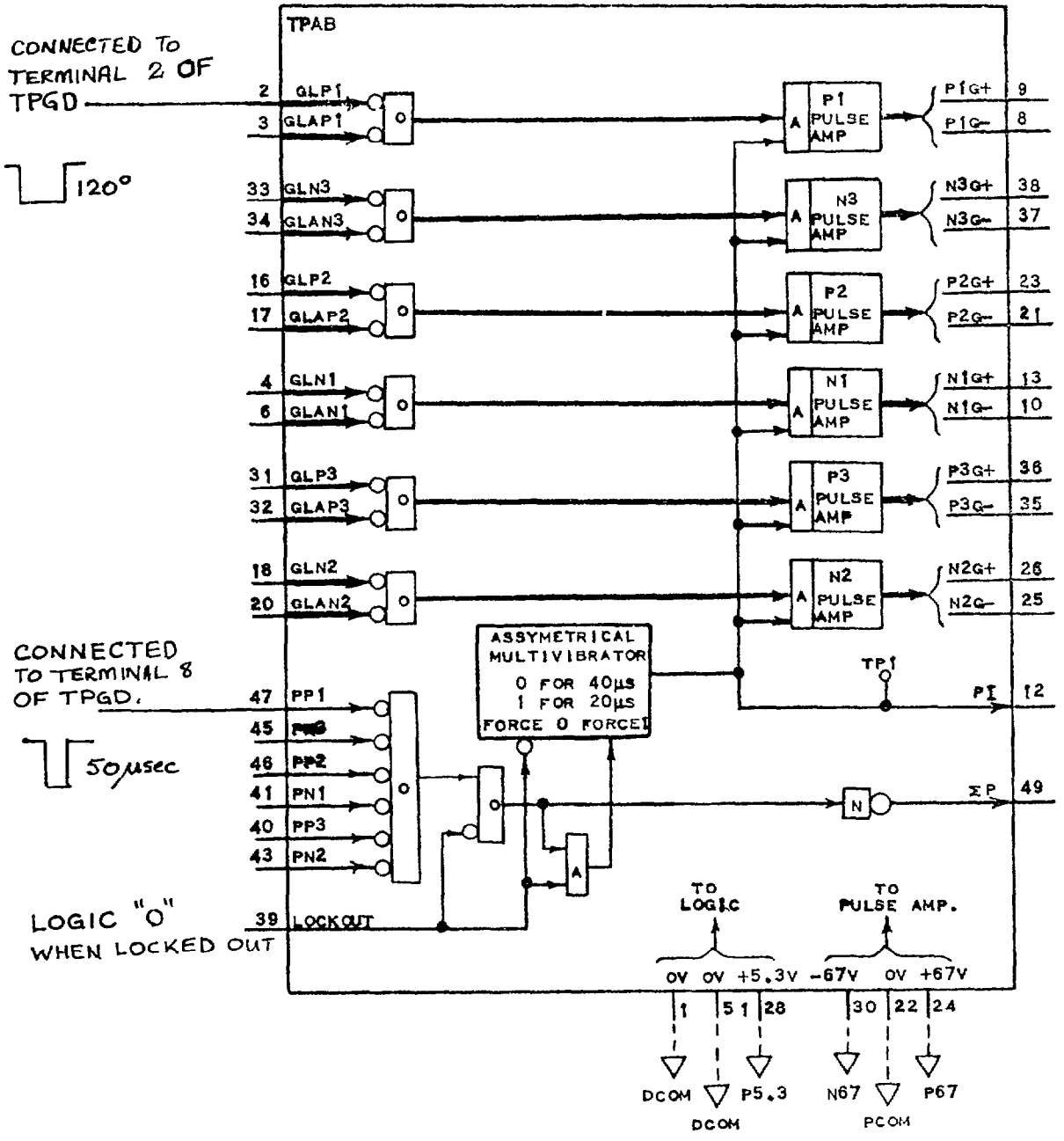


Figure 17

SILTROL I START-UP PROCEDURES

All Siltrol I units are completely factory tested and the line-ups are shipped as one unit. Therefore, field testing should be reduced to a simple check list and fine-tune with the actual drives. The purpose of these notes is to serve as a sequential guide for installation and checkout of the Siltrol equipments.

Precautions

First, the following precautionary advice is included to forewarn the tester against committing procedures which might tend to nullify important protective relationships that are designed into the system.

A. For those rectifiers fed from low voltage primary where the rectifier transformer is approximately 1:1 in turns ratio, DO NOT operate the rectifier directly from the primary source (bypassing the transformer). The transformer is designed with a specific value of reactance to limit available short-circuit current. Without this reactance the armature supply could be subject to loss of thyristors.

B. DO NOT megger control modules and circuits where solid state electronic devices are present, as they can be damaged easily.

   When meggering rectifier power circuits, short all busses together to avoid impressing potential across rectifying devices unnecessarily. Hi-potting of rectifier power circuits is not considered advisable or necessary.

C. Use extreme care and caution when working with or around the rectifier power circuits and modules. They are by nature high capacity, low impedance paths, and a destructive arcing fault could be precipitated by careless procedure. For example connecting scope to d-c bus without proper probes is dangerous. Refer to oscilloscope techniques.

D. DO NOT operate the rectifier with Suppression circuit cards or a-c CT circuits opened. During inspection or trouble-shooting procedures, circuits may be systematically opened or isolated as logical maneuvers, but care must be taken not to nullify the Suppression function, or leave CT's open circuited.

E. To maintain the correct cooling capacity in the Siltrol equipment, it is imperative that none of the rear covers be removed prior to operating the equipment.

F. DO NOT operate the rectifier full-reversing on a regenerative load with the Reversing Logic or Suppression circuits disabled.

Oscilloscope Techniques

Observations of voltage and current waveforms in power rectifier circuits require extra care in selection and use of oscilloscope equipment. The make-up of the rectifier bridge circuit, whether grounded or not, produces large voltages to ground at nearly all points. In observing voltages or currents within the rectifier, differential signal circuits in the oscilloscope are used to reject these common mode voltages and to display the desired signal correctly.

In addition to more accurate observations, differential signal circuits are also safer to use since the oscilloscope chassis is grounded. The oscilliscope equipment should be provided with calibrated vertical deflection system to allow interpretation of observations. Since the oscilloscope equipment must accommodate the highest voltage in the rectifier, voltage probes are used to extend the voltage rating. These probes should be matched in rating and performance for use in differential observations. "Tektronic" Type 453 portable scope with suitable voltage and current probes is recommended to examine rectifier wave forms. For systems up to 600 volts use Type P6006-10X voltage probes. Type P6023 probes are often very useful to improve rejection of voltage signals to grounds for more accurate presentation of desired waveforms. To examine thyristor gate currents use Type P6021 current probe.

The sweep system should be calibrated in sweep speed, and should preferably be of a triggered type to allow reliable display of the signal. If waveforms are to be recorded on film, the sweep generator should also include a single sweep feature. Type 453 contains all those features.

The oscilloscope equipment suggested must be correctly adjusted to produce accurate results. First the pre-amplifier is adjusted for d-c balance (set AC-GND-DC switch to GND, set sweep mode switches to auto trig) by varying the step attenuator control until no trace of motion is seen as the variable vertical calibration control is moved from 20MV to 5MV. Both voltage probes must be adjusted for correct frequency compensation using the calibrator as a square wave standard and adjusting the a-c coarse compensation control in the termination box of the probe.

Voltage calibration of the probe and pre-amplifier is adjusted by the gain adjustment using the calibrator signal as a standard. The above operations are done with one signal circuit at a time. The scope is changed to display the difference between channel 1 and 2 by setting mode switch on ADD and inverting the channel 2 signal. Both probes are connected to the calibrator. Set calibrator on 1V. Turn trigger from normal to channel. Adjust gain control of channel 2 to obtain straight line with volts/div control set on 10MV per division for 10x probes. Next the pre-amplifier is changed to differential connection and both probes are connected to the calibrator. Adjust the d-c attenuator control and a-c compensation fine adjustment for the minimum signal display. In final adjustment, common mode rejection can be checked by connecting both probes to the same circuit point to be tested. The display should remain a straight and stationary line.

Siltrol I Checklist -- Full Reversing

1. Perform preliminary inspection as applicable to any new equipment. Check proper assembly and connection, both mechanical and electrical; loose, missing or damaged parts.

2. A "Siltrol Power Converter Design Variable Data Sheet" is supplied as part of the System Drive Elementary for each drive. Each System Drive Elementary also has a sheet showing the Siltrol Arm. Supply in block form with "application variable information." Using these sheets and the Siltrol Elem. check the following:

   A. Inspect Primary Breaker (Refer to GEH-2021C for AK type breakers)

   B. On Cards check:

      I Values of R12, R13, R14 and R15 on CCCA card, slot BOD.

      II All jumpers per drive elementary on CORD card, slot BOH.

3. Drive Magnetics and Drive Regulator

   A. Check out drive magnetics (line panel, relay panel, etc.) using system elementary. If drive has logic check out using system elementary.

   B. Ensure that Siltrol suicide circuit operates and is connected to Siltrol.

   C. Disable drive regulator.

4. Siltrol Tests

   A. Disconnect the power bridge from the rectifier transformer secondary.

   B. Connect resistive load to Siltrol output. (Suggest 10 ohms, 5KW).

   C. Select magnetics so that when suicide signal is lifted the M contactor closes and connects the load.

   D. Apply a suicide signal (i.e. suicide the Siltrol)

   E. Close a-c breaker and check the following:

      I Cooling blower motor should start and circulate air through the bridge from the front and then over the transformer. Motor rotation is correct when the fan runs with a high noise level and stops rapidly on removal of power. The blower air flow switch should close and operate relay BL.

      II Transformer overtemperature relay should energize.

III Verify presence of power with a multimeter at the following points:

+67V (+7V -4V) BOI	Pin 24	ACOM	BOI2
-67V (+7V -4V) BOI	Pin 30	ACOM	BOI2
+5.3V(+.5V-.5V) BOI	Pin 28	DCOM	BOI1

IV Ensure phase rotation is L_1, L_2, L_3, using a phase rotation meter or scope.

F. Connect a $\pm$ 15V d-c supply to REF 1 (Note: It may be convenient to use the drive regulator output). With zero signal check the presence of pulses at each thyristor gate using a current probe. Pulses should be present for package B only. Connect a 10K resistor across the operational amplifier, remove the suicide signal and swing REF 1 positive and negative and check that pulses advance smoothly and appear for one package at a time. Apply suicide signal and remove the 10K resistor.

G. Check the I_A and I_B logic operation and first and second overcurrent settings by injecting a milliamp signal into I_A (OI16) and I_B (OI18). Check the requisition elementary data sheet for milliamp settings for the overcurrent levels. If a suitable milliamp source is not available it may be necessary to remove the burden resistor R12, R13, R14, R15 on card CCCA and inject a suitable voltage signal to do the tests.

REPLACE THE BURDEN RESISTOR AFTER TEST.

H. Apply suicide signal and open Siltrol breaker. Reconnect the power bridge.

I. Reclose breaker and remove suicide signal. Check for zero output voltage. Note: The voltage regulator loop is now closed.

J. Vary REF 1 to $\pm$ 15 volts and observe d-c output voltage. The waveform produced should be a sawtooth with 60° spacing.

K. Set REF 1 to 7.5V and check that the d-c output is 50% of rated. VIA should be -7.5 volts.

L. Check operational amplifier output with a scope. The waveform should be a smooth 360 Hz ripple (300 Hz for 50 Hz power system) approximately 5V peak to peak in the continuous current region.

Unbalanced ripple or deviation from approximately 5V peak to peak indicates regulator instability.

M. Check that I_L feedback is correct in sense and magnitude.

N. Vary input from +VE to -VE and observe d-c output voltage with a scope for any irregularity.

O. Disconnect the Siltrol from the resistor and connect to the drive. The Siltrol tests are now complete except for a check of first and second level overcurrents with actual current. It is not recommended that excessive currents be circulated to check trip points, but actual current should be checked against I_A and I_B to check operation of the CT's.

Siltrol Checklist - Non-reversing

1- 4E. Perform these steps as for reversing drives.

4F. Connect a -15V DC variable supply to REF 1 (Note - It may be convenient to use the drive regulator circuit.) If the drive has a voltage regulator connect a 10K resistor across the operational amplifier. With zero signal check that pulses are present on all thyristor gates. Lift suicide and swing REF 1 negative and check that pulses advance smoothly. Apply suicide signal and remove 10K resistor.

G. Check I_A first and second overcurrent levels as for reversing drive.

H. Apply suicide signal and open the Siltrol breaker. Reconnect the power bridge.

I. Reclose breaker and remove suicide signal. Check for zero output volts and current. Note: If the drive has a Siltrol voltage regulator the loop is now closed.

J. Vary reference from zero to -15V and observe d-c output voltage. The waveform produced should be a saw tooth with 60° spacing. The waveform should remain evenly balanced in amplitude and spacing.

K. If drive has voltage regulator set reference to -7.5V and check that the d-c output voltage is 50% of rated. VIA output should be 7.5 volts.

L. If the drive has a voltage regulator the operational amplifier output waveform should be a smooth 360 Hz ripple (300 Hz for 50 Hz power system) approximately 5V peak to peak in the continuous current region.

Unbalanced ripple or deviation from approximately 5V peak to peak indicates regulator instability.

M. Check that the I_L feedback is correct in sense and magnitude.

N. Disconnect the Siltrol from the resistor and connect to the drive. The Siltrol tests are now complete except for a check of first and second level overcurrents with actual current. It is not recommended that excessive current be circulated to check trip points, but actual current should be checked against I_A to check operation of the CT's.